# Evaluation-Guided Domain Randomization: Certificate Margins and Controller Updates

**Shimin Chen** CSHIMIN@ENGINEERING.UPENN.EDU
*Department of Electrical and Systems Engineering, University of Pennsylvania*

## Abstract

Evaluation-guided domain randomization targets training conditions using a controller's observed weaknesses, but improvement on selected conditions can cause regression elsewhere. We propose a theory-focused study of a small linear-system family: specialize trajectory-based certificate learning to quadratic Lyapunov candidates, then analyze one controller update constrained by a verified decrease margin. Cost-sensitivity bounds and counterexamples will separate stability preservation from performance-coverage improvement. Small numerical experiments support the analysis; cost-region expansion or a local nonlinear extension is an ambitious additional outcome.

## 1. Motivation

Domain randomization trains a shared controller across varying plants and initial conditions. Evaluation can identify conditions near a performance threshold and guide subsequent training, but proximity to that threshold need not imply useful training signal. We study what can be certified before an update and preserved afterward. The L4DC connection is the interaction between statistical generalization of learned certificates and feedback-dependent trajectory changes when the controller is updated.

## 2. Problem Formulation

Consider a known finite plant family with full-state feedback, $x_{t+1} = A(\theta)x_t + B(\theta)u_t$, $u_t = -Kx_t$, and $G_K(\theta) = A(\theta) - B(\theta)K$. The shared gain $K$ is frozen during evaluation; $\theta$ stays fixed within each trajectory. The initial model is deterministic, continuous-action, and unconstrained. Known matrices permit exact verification alongside certificate fitting from sampled transitions. The retention family must admit a common quadratic certificate; this is an explicit restriction beyond individual plant stability.

The method connects three feedback loops: **control** maps states to actions and next states; **evaluation** maps chosen conditions to trajectory-based certificate and cost evidence; **improvement** uses that evidence to choose a training distribution, proposes an update, checks its margin, and freezes the accepted controller for fresh evaluation.

For fixed $Q, R \succeq 0$, let $J_{H,K}(\theta, x) = \sum_{t=0}^{H-1}(x_t^\top Q x_t + u_t^\top R u_t) = x^\top P_H(K, \theta)x$, with $x_0 = x$. Define $S_K^J = \{(\theta, x) : \|x\|_2 \le r,\ J_{H,K}(\theta, x) \le h\}$. Coverage is $\mu(S_K^J)$ under a fixed reference probability measure $\mu$; the training distribution $q$ may change. Net coverage change equals gained mass minus lost mass. The cost matrix $P_H$ is distinct from a Lyapunov candidate $P$, and finite-horizon acceptability does not establish stability.

**Research questions.** What generalizes when a quadratic certificate is fitted to trajectories of a frozen controller? Which verified margins survive an evaluation-guided update? When can targeted training reduce its objective yet lose acceptable coverage?

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## 3. Goals and Approach

**Stage 1: Course foundation.** We will specialize Boffi et al. (2021) to discrete time and $V_P(x) = x^\top P x$, with fixed $0 < aI \preceq P \preceq bI$. For frozen $K$, collect $H$ transitions per independent plant/initial-state pair from a fixed distribution. A bounded margin loss measures violations of $V_P(x_{t+1}) - V_P(x_t) \le -\eta\|x_t\|_2^2$, $\eta > 0$. This Lyapunov condition requires energy to fall away from zero, with decrease vanishing at the equilibrium. If valid for all states, it gives $V_P(x_{t+1}) \le (1 - \eta/b)V_P(x_t)$ and exponential convergence (Chapter 3, Lemma 3.11). A finite-cover uniform bound (Chapter 5) will control violation risk, treating whole trajectories as independent samples; it does not prove all-state stability or claim the source paper's strongest rate. Separately, known matrices verify $G_K(\theta)^\top P G_K(\theta) - P \preceq -\eta_\theta I$, $\eta_\theta > 0$, equivalent to all-state decrease on each retained plant. Report failed verification.

**Stage 2: Main theoretical investigation.** Hold verified $P$ fixed. For $K' = K + \Delta K$, write $G = G_K(\theta)$ and $d_\theta = \|B(\theta)\Delta K\|_2$. Expanding the quadratic form gives the sufficient preservation test

$$2\|G^\top P\|_2 d_\theta + \|P\|_2 d_\theta^2 < \eta_\theta.$$

We will connect this test to one concrete update: evaluate old costs on a fixed grid, mix a target near $J_{H,K} = h$ with fixed background sampling, and take an exact-gradient step for $F_q(K) = \mathbb{E}_q J_{H,K}$. Hold $q$ fixed during backtracking and require training descent and a positive retained margin on every retention plant; a capped unsuccessful search leaves $K$ unchanged. This preserves verified stability, not the cost-acceptable set. We will also derive an explicit bound $\|P_H(K', \theta) - P_H(K, \theta)\|_2 \le L_H(\theta)\|\Delta K\|_2$, including the changed input penalty. With $b_J(\theta) = r^2 L_H(\theta)\|\Delta K\|_2$, losses lie in $h - b_J(\theta) < J_{H,K} \le h$. A targeted-gradient counterexample will show that training descent and stability can coexist with reduced coverage. These elementary results and their limits, rather than a general PPO theorem, form the required analysis.

**Stage 3: Ambitious extension.** Seek net expansion of $S_K^J$ under a specified update by establishing gains that outweigh losses. Alternatively, study a local nonlinear extension with an explicit remainder bound and invariant neighborhood. Choose one extension; Stages 1–2 remain deliverables without it. One outer update is sufficient.

**Supporting experiments.** Small linear examples will compare exact and bounded drift, certificate verification, and gains/losses for uniform versus cost-boundary targets under the same update restriction. CartPole is an optional nonlinear illustration: with a 90° angle cutoff and other termination rules retained, recovery requires 500-step survival and final-100-step RMS angle at most 5°. Its discrete actions and large-angle behavior do not inherit the linear theorem. Recovery claims require fresh episodes and simultaneous bounds (Howard et al., 2021). Retain raw trajectories, configurations, and seeds.

## 4. Related Work

Boffi et al. (2021) study certificate generalization; Zhang et al. (2022) extend this to robust certificates under incremental stability assumptions. Domain-randomized LQR (Fujinami et al., 2025) motivates shared-gain learning. Reverse curricula (Florensa et al., 2017) precede performance-guided sampling, while Berkenkamp et al. (2017) study Lyapunov-based safe-region expansion. These results motivate our analysis but do not imply coverage expansion for the proposed update.
*AI assistance:* Codex assisted with drafting and preliminary derivations.

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